My Final College Paper

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I want to thank a few people.

Preface

This is an example of a thesis setup to use the reed thesis document class.

List of Abbreviations

You can always change the way your abbreviations are formatted. Play around with it yourself, use tables, or come to CUS if you'd like to change the way it looks. You can also completely remove this chapter if you have no need for a list of abbreviations. Here is an example of what this could look like:

ABC	American Broadcasting Company
CBS	Columbia Broadcasting System
CDC	Center for Disease Control
CIA	Central Intelligence Agency
CLBR	Center for Life Beyond Reed
CUS	Computer User Services
FBI	Federal Bureau of Investigation
NBC	National Broadcasting Corporation

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Abstract

The preface pretty much says it all.

Dedication

You can have a dedication here if you wish.

Introduction

0.1 Thesis of the Thesis

Aside from visual presentation of one's body, the sound of a person's voice is among the most strongly gendered aspects of behavior. Trans people know this quite intimately, as our speech is highly scrutinized by those who – consciously or unconsciously – seek to enforce the gender binary. In my thesis, I plan to investigate how this knowledge manifests in English-speaking transfeminine speakers' linguistic practice with respect to an aspect of pronunciation which is often highly associated with gender in western hegemonic gender normativity – the pitch of the /s/ sound – through a comparison of their speech in a conversation with a cis person versus in a conversation with a trans person.

Chapter 1

Literature Review

1.1 Brief Literature Review

The above claim that trans speech is often more scrutinized by cisgender people for gender conformity, that trans people are aware of this, and that their knowledge may manifest in linguistic practice is based upon existing findings in the field of transgender linguistics. Investigating transgender knowledge, Heffernan (2023) finds that transfeminine speakers characterize a 'cisgender listening subject' in their metalinguistic commentaries which casts negative judgements when linguistic gender norms are not followed, and that transfeminine speakers expressed varied ways of negotiating this, from accomodating these expectations by a change in linguistic behavior towards conformance to asserting gender to be self-determined, resisting these judgements, and talking as one would prefer. Jessee and Calder (2025) emperically confirm that transgender speech is indeed subject this greater gender scrutiny than cisgender speech, finding that cisgender listeners apply stricter gendered standards to speech when they are told the speaker is a transwoman than when told the speaker is a ciswoman. Although Parnell-Mooney (2019) predates these more recent findings, he presents a design very similar to my own but with a focus on transmasculine and nonbinary speakers, in which it was found that these speakers do in fact pronounce /s/ with different pitch when talking to a trans person versus a cis person, but that variation was did not pattern across speakers.

My choice to focus on transfeminine speakers is motivated by three reasons: the relative scarcity of sociolinguistic research which investigates transfeminine speech on the phonetic level, the emperical gap left by left by the exclusion of transfeminine speakers from Parnell-Mooney (2019), and the contributions to sociolinguistic theory, especially existing questions in the scholarship on /s/, that may arise by investigating transfeminine speech. The scarcity of sociophonetic research on transfeminine speakers has been noted by Zimman (2020); in a review of the place of trans people (and other groups which fall outside of the gender binary) in the language and sex, gender, and sexuality literature, they find that in the Global South, research has primarily focused on third and fourth-gender groups which

bear similarities to transfeminine people, and in the Global north it has leaned towards transmasculine people, "leaving trans feminine people in the Global North and trans masculine people in the Global South underrepresented." As I turn to the reasoning behind my choice to look at /s/, I will demonstrate this gap and discuss the ways in which a study of transfeminine interlocutor-oriented style-shifting (the sociolinguistics term for changes in an individual's speech depending on who they're talking with, see eg. Rickford and Mcnair-Knox 1994) could contribute new insights on existing theoretical questions raised by prior work on /s/.

Termed 'sexuality's most iconic variable' by Calder (2020), /s/ has become one of the most well-studied sites of socially-relevant variation in speech (called 'variables' in sociolinguistics) in the field of language and sex, gender, and sexuality. This scholarship has primarily oriented itself around finding differences in /s/ pronunciation and perception between sex, gender, and sexuality groups, individuals, and contexts, and attempting to explain those differences. The most commonly studied aspect of /s/ is frontness, an analouge of /s/ pitch; a fronter tongue position produces a higher-pitch (see Fuchs and Toda 2010). Specifically, studies consistently find that women produce significantly fronter /s/ than men, and that gay men produce significantly fronter /s/ than straight men (eg. Podesva and Van Hofwegen 2015; Hazenberg 2012; Flipsen et al. 1999; also reviewed in Calder 2020), and that /s/ is linked to women, gayness, and femininity by listeners (eg. Campbell-Kibler 2011; Maegaard and Pharao 2015).

Using the framework of indexicality, by which linguistic features come to index ('point to') social meanings, scholars have considered the social meaning of fronted /s/ to be femininity. That /s/ comes to be associated with gay masculinity is understood by Calder (2019) to be an example of a meaning of a higher indexical order – a notion developed by Silverstein (2003) by which a linguistic feature comes to have new meanings through the process of adaptation to a new usage context. In this case, the use of fronted /s/ by gay men adapts its original meaning, femininity, to its new context, gayness. This structure of meanings comes to be through the imagining of gay men as more feminine, or more woman-like, than straight men. Calder (2019) sees this as an example of the ideological process Irvine and Gal (2000) term fractal recursivity by which a divison of society gets projected inside of itself– gay men become "men who are women" through fractal recursivity, and thus, at their use of a 'woman's' linguistic feature, new indexical order of /s/ emerges, going from a link from fronter /s/ to femininity to a link from fronter /s/ in men to gayness.

Although this framework of meanings works very well to describe the data on white, middle-class, cisgender westerners, Calder (2021) points out that this structure, although 'real' (that is, truly present and salient for the people it describes), does not work for those outside of this hegemonic group. If fronted /s/ were simply a feminine and gay-male index, trans people would be expected to pattern just the same as cis people. The fact that they do not (as seen in, eg. Hazenberg 2012; Podesva and Van Hofwegen 2015; Zimman 2013) also cannot be explained as simply another level of fractal recursivity; unlike sexuality identity, which can construe itself ideologically as a layer over and above gender and sex in the fractal

recursive tree, this is untenable for transness, which is unavoidably present at all levels. Similarly, that these patterns do not hold outside of the sphere of whiteness (see eg. J. Calder and King 2022; Jessee and Calder 2025; Maegaard and Pharao 2015) exposes the above understanding of the meaning structure of fronted /s/ as limited and incomplete.

Using the concept of the indexical field, developed by Eckert (2008), which refers to the set of all meanings a linguistic feature can have, depending on context, Calder (2021) argues that the majority of the research on /s/ which does not fall into the more recent fields of raciolinguistics and trans linguistics, tacitly normalizes the white, cis, western, middle-class subset of the much more expansive indexical field. They argue that an account of how sex, gender, sexuality, (and race) are produced through language is incomplete without investigations of "speaking and listening subjects outside of White and cisgender populations." (43) This further motivates my study of transfeminine speakers; I hope to find how they navigate this hegemonic meaning structure, and what meanings they use /s/ frontness to create. A more recent theorization of indexicality by Barrett and Hall (2024) suggests that speakers and listeners bring their own indexical inventories to an interaction, and differences between these, or intentional exertion of power over another, can result in instances of misrecognition, by which the intended meaning of a n indexical behavior is misinterpreted. By looking at how transfeminine speakers navigate the possibility of misrecognition (eg. a fronted /s/ making them sound like a gay man, or a backed /s/ making them sound like a man rather than a lesbian), I hope evaluate this theoretical position.

Furthermore, I hope to raise questions about the status of 'meaning' as the function of linguistic behavior. In the same way as literary theorists and feminists have attempted to move analysis in their fields away from the search for a meaning and towards an understanding of doing (Deleuze and Guattari 1987; Butler 1990; see also Livia and Hall 1997), I feel it is possible that sociolinguistics needs to take this turn; my study of how transfeminine speakers deal with those who hope to impose meaning on their speech is an attempt to address this question: do they engage in the game of meaning-making, either by developing their own meanings, taking up up hegemonic meanings, or strategizing around the indexical inventory of their interlocutor, or do they reject 'meaning' entirely, instead focusing on what their speech can do for them within whichever context they find themselves in. Are these theorizations even properly distinct?

Persuant to both the emperical gaps around transfeminine speech, and the theoretical questions just raised, I plan to investigate the **research question**: How do transfeminine speakers' pronunciations of English /s/ compare in speech targeted to a cisgender interlocutor versus in speech targeted to a transgender interlocutor with respect to frontness, and how does this vary across speakers of differing social backgrounds and identities?

Note: The below sections are a rough outline for my fuller lit review. Content taken from the brief lit review above will eventually be situated into sections something like these

1.2 Queer Linguistics, Trans Linguistics

1.3 The Variable: /s/

1.4 Indexicality

1.5 Intersectionality, Trans linguistics, and Raciolinguistics

1.6 Performativity

1.7 Feminism, Queer Theory, Literary theory, and Trans Studies

Chapter 2

Methodology

I plan to investigate this by gathering data on /s/ pitch used in conversations between transfeminine speakers and a cis woman interlocutor, and transfeminine speakers and a trans woman interlocutor via acoustic analysis. I plan to analyse this data quantitavely using statistical methods, as well as qualitatively using information gathered in interviews between myself and the speakers, and, as time permits, discourse analysis.

2.1 Participants

I plan to investigate this question through the recruitment of 6 participants from the Reed College student body, 2 of which will be my interlocutors – a cis woman and a trans woman. The other 4 will be transfeminine people – my speakers – ideally diverse in their identification with respect to the binary gender 'woman'.

2.2 Procedure of Recording Meetings

After recruitment, meetings will be scheduled such that each speaker has a short conversation (about 20min each) with the cis interlocutor, then trans interlocutor, then myself. Interlocutors will additionally meet with me beforehand to tell me about their background and recieve orientation to what will be expected of them should they provide informed consent.

Recording meetings with speakers will all take place in the Lab of Linguistics, and I will be present just before each conversation to orient participants to the experiment, confirm informed consent, and set up recording equipment. Prior to each of the conversations between the speakers and interlocutors, the structure of the conversation will be explained. The conversation will have two modules: first, the pair will be asked to collaboratively complete a crossword puzzle, and second they will be asked to spend the rest of the available time getting to know each other and discussing experiences or interests they find they have in common. The purpose of the first module is to act as an 'icebreaker' to get the speakers

talking, and to provide a conversational context in which the personal identities of the conversants are not forefronted. The purpose of the second module is to provide the opposite conversational context, in which personal identity is forefronted. This design is informed by Holmes-Elliott and Levon (2017) and Rickford and Mcnair-Knox (1994), in which it was found that style-shifting depends on topic (in my design, task-oriented vs. identity oriented), and stance-taking (in my design, oppositional, [eg. disputing a crossword answer] vs relational [trying to find commonalities and relate to one another]).

The recordings of the speakers within these meetings will be the subject of acoustic analysis. The recordings of participants (interlocutors and speakers) meetings with myself will be recorded also, mainly to be available to reference back to in favor of fallible memory or notes. In these meetings, I will get to know the participants, aiming to understand their sex, gender, and sexuality identities, especially the speakers' transgender journeys, backgrounds, experiences, and metalinguistic orientations. Within these interviews, participants and I would settle on a system of coding sex, gender, and sexuality identity, racial identity, as well other aspects of background (eg. years since transition, english monolingual vs. other languages, etc) which is comfortable for them, and these codings will be operationalized as variables the statistical analysis.

2.3 Analysis

In acoustic analysis, I will measure /s/ pitch by the metric of center of gravity (COG), following the large majority of acoustic research on /s/ (see eg. Fuchs and Toda 2010; Podesva and Van Hofwegen 2015; Hazenberg 2012). This would be achieved first through orthographic transcription of breath groups in Praat, forced alignment of segments in DARLA, and measurement of /s/ token COG using a Praat script.

From there, I would quantitavely analyse the relationships between the identity and background variables (as decided through the interview process), whether the interlocutor was trans, and the /s/ realizations within the associated recording. I would use both descriptive and inferential statistical methods to understand the patterns in the data. Inferrential statistics would take the form of a linear mixed effects model in R.

For the qualitative analysis, I would consider the information learned about each participant in my interviews with them, and apply this knowledge to form an understanding of their behavior with respect to /s/. Additionally, I may ask for consent to show certain transcribed portions of the speech for the purposes of discourse analysis.

2.4 Funding

Estimation and breakdown of participants' hours:

- 4 'Speaker' participants will each be present for one 1h15m meeting (setup + 3 approx. 20min recording sessions)
- 2 'Interlocutor' participants will each be present for and one 30min meeting (for me to get to know them and orient them to the study) and four 35min meetings (setup + recording session with one 'speaker' participant), totaling 2h50m per 'interlocutor' participant.
- 'Speaker' participants time would total to 4h45m, and 'Interlocutor' participants time would total to 5h40m.
- **Participants' time would total 10h25m**.

Setting a payrate of $18 per hour, this would require an estimated total participants' compensation of $187.50. No further funding needs are expected, as recording equipment will be provided by the Lab of Linguistics, and free software will be used for data collection and analysis.

Chapter 3

Results

3.1 By Independent Variable

3.2 By Dependent Variable

Chapter 4

Discussion

4.1 Comparison with Prior Results

Conclusion

5.2 Directions for Future Research

Appendix A

The First Appendix

Appendix B

The Second Appendix, for Fun

Bibliography

Austin: How to do things with words

J. L. Austin. *How to do things with words*. The William James lectures 1955. Oxford: Clarendon Press, 1962.

Annotations: In this book, the concept of the linguistic performative is developed. As the genesis of performativity within the field of linguistics, only later reunited to its life in social theory, this is a foundational theoretical text to performativity theory, which I will make use of in my analysis.

Barrett et al.: Sexuality Discourses

Rusty Barrett and Kira Hall. "Sexuality Discourses: Indexical Misrecognition and the Politics of Sex". In: *Annual Review of Anthropology* 53 (Volume 53, 2024 Oct. 21, 2024). Publisher: Annual Reviews, pp. 127–146. ISSN: 0084-6570, 1545-4290. DOI: 10.1146/annurev-anthro-052721-092531. URL: https://www.annualreviews.org/content/journals/10.1146/annurev-anthro-052721-092531 (visited on 09/13/2025).

Annotations: I hope to engage with this theory which relativizes indexicality through the notion of individuals having their own indexical inventories (subsets of the whole indexical fields of the variables they are aware of) and considers misrecognition of indexical meanings a place for power to enter the indexicality structure, specifically whether it is consonant with, or in tension against, performativity theory and queer theory.

Butler: Gender Trouble

Judith Butler. *Gender Trouble: Feminism and the Subversion of Identity*. 0th ed. New York: Routledge, 1990. 272 pp. ISBN: 978-0-203-82497-9. DOI: 10.4324/9780203824979. URL: https://www.taylorfrancis.com/books/9781136783241 (visited on 04/08/2025).

Annotations: A foundational text to third-wave feminism, I take up Butler's theorization of gender as performative for my analysis.

Calder: From "Gay Lisp" to "Fierce Queen"

J Calder. "From "Gay Lisp" to "Fierce Queen": The Sociophonetics of Sexuality's

Most Iconic Variable". In: *The Oxford Handbook of Language and Sexuality*. Ed. by Kira Hall and Rusty Barrett. 1st ed. Oxford University Press, Dec. 15, 2020. ISBN: 978-0-19-021292-6 978-0-19-021294-0. DOI: 10.1093/oxfordhb/9780190212926.013.49. URL: https://academic.oup.com/edited-volume/42645/chapter/358158508 (visited on 04/05/2025).

Annotations: Part of Calder's work on SoMa drag queens, I use this paper for its review of prior literature on /s/, and engage with its conceptualization of /s/'s iconicity.

Calder: From Sissy to Sickening

J Calder. "From Sissy to Sickening: The Indexical Landscape of /s/ in SoMa, San Francisco". In: *Journal of Linguistic Anthropology* 29.3 (2019). _eprint: https://onlinelibrary.wiley.com/d pp. 332–358. ISSN: 1548-1395. DOI: 10.1111/jola.12218. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/jola.12218 (visited on 04/04/2025).

Annotations: Part of Calder's work on SoMa drag queens, I use this paper for its discussion of the indexicality of /s/: it theorizes meanings of /s/ as belonging to indexical orders which form along strata of fractal recursivity, and raises the insight that the presentation of the body is part of the stylistic context which determines at which indexical order /s/ behavior will be interpreted through.

Calder: Whose indexical field is it?: the role of community epistemology in indexing social meaning

J Calder. "Whose indexical field is it?: the role of community epistemology in indexing social meaning". In: *Proceedings of the 20th Meeting of the Texas Linguistics Society*. Ed. by Caitie Coons et al. Texas: Texas Linguistics Society, 2021, pp. 39–55. URL: https://tls.ling.utexas.edu/2021/TLS_2021_Proceedings.pdf.

Annotations: This conference paper brings the concept of the indexical field (see Eckert 2008) into the study of /s/, as well as language and gender more broadly, especially highlighting the importance of intersectional approaches which do not implicitly assume a white, cisgender, heterosexual, western, middle class, etc. indexical field to be a culturally-independent reality. I use this as part of the motivation of my study's focus on transgender speakers.

J. Calder et al.: Whose gendered voices matter?

J. Calder and Sharese King. "Whose gendered voices matter?: Race and gender in the articulation of /s/ in Bakersfield, California". In: *Journal of Sociolinguistics* 26.5 (2022). _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/josl.12584, pp. 604–623. ISSN: 1467-9841. DOI: 10.1111/josl.12584. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/josl.12584 (visited on 04/05/2025).

Annotations: I use this paper as an example of a raciolinguistic approach to the study of /s/ which attempts to decenter the white indexical field, and illustrates the whiteness of the traditional sociolinguistics structure of /s/ meanings.

Campbell-Kibler: Intersecting Variables and Perceived Sexual Orientation in Men

Kathryn Campbell-Kibler. "Intersecting Variables and Perceived Sexual Orientation in Men". In: *American Speech* 86.1 (Feb. 1, 2011), pp. 52–68. ISSN: 0003-1283. DOI: 10.1215/00031283-1277510. URL: https://doi.org/10.1215/00031283-1277510 (visited on 04/05/2025).

Annotations: A perception study of /s/ which finds social meanings of "less masculine, more gay, and less competent" arise for fronted /s/ within certain stylistic contexts of an (otherwise) male-sounding voice. I use this as one of the baseline studies of the hegemonic meanings of /s/.

Deleuze et al.: November 20, 1923: Postulates of Linguistics

Gilles Deleuze and Félix Guattari. "November 20, 1923: Postulates of Linguistics". In: *A thousand plateaus: capitalism and schizophrenia*. Trans. by Brian Massumi. Minneapolis, MN: University of Minnesota Press, 1987, pp. 75–110. ISBN: 978-0-8166-1401-1.

Annotations: In this chapter, poststructuralist philosophers Deleuze and Guattari present a critique of the structuralism of linguistics, as well as a critique of the foregrounding of meaning, and highlight that doing always preceeds and overpowers meaning. I specifically use their notion of the order-word as connected to Austinian (see Austin 1962) performativity.

Eckert: Variation and the indexical field

Penelope Eckert. "Variation and the indexical field". In: *Journal of Sociolinguistics* 12.4 (2008). _eprint: https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1467-9841.2008.0037 pp. 453–476. ISSN: 1467-9841. DOI: 10.1111/j.1467-9841.2008.00374.x. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/j.1467-9841.2008.00374.x (visited on 09/14/2025).

Annotations: I take up the concept of the indexical field, a way of theorizing the different social meanings a sociolinguistic variable can index, that this paper introduces.

Flipsen et al.: Acoustic Characteristics of /s/ in Adolescents

Peter Flipsen et al. "Acoustic Characteristics of /s/ in Adolescents". In: *Journal of speech, language, and hearing research* 42.3 (1999). Place: United States Publisher: ASHA, pp. 663–677. ISSN: 1092-4388. DOI: 10.1044/jslhr.4203.663.

Annotations: I use this paper as a reference on "typical" /s/ pronunciations in the cishet white western normative english sphere, esp. for its review of 21 prior studies on /s/.

Flores et al.: Undoing Appropriateness

Nelson Flores and Jonathan Rosa. "Undoing Appropriateness: Raciolinguistic Ideologies and Language Diversity in Education". In: *Harvard Educational Review* 85.2 (June 1, 2015), pp. 149–171. ISSN: 1943-5045, 0017-8055. DOI: 10.17763/0017-8055.85.2.149. URL: https://meridian.allenpress.com/her/article/85/2/149/32176/Undoing-Appropriateness-Raciolinguistic-Ideologies (visited on 09/09/2025).

Annotations: This study theorizes the white listening subject as highly relevant to racially minoritized people's linguistic experience. I hope to bring it into my analysis of how race is working within my study, as well as to emphasize the importance of making culturally-specific claims.

Fuchs et al.: Do differences in male versus female /s/ reflect biological or sociophonetic factors?

Susanne Fuchs and Martine Toda. "Do differences in male versus female /s/ reflect biological or sociophonetic factors?" In: *Turbulent Sounds: An Interdisciplinary Guide*. Ed. by Susanne Fuchs, Martine Toda, and Marzena Zygis. De Gruyter Mouton, Apr. 29, 2010, pp. 281–302. ISBN: 978-3-11-022658-4. URL: https://www.degruyterbrill.com/document/doi/10.1515/9783110226584.281/html (visited on 09/26/2025).

Annotations: An electropalatographic study of /s/ which determines that differences in /s/ pitch are caused by differences in tongue position, and considers whether variation in /s/ is socially constructed, or a result of so-called 'biological sex'. I use this paper for its articulatory findings.

Hazenberg: Language and identity practice

Evan Nicholas Leo Hazenberg. "Language and identity practice : a sociolinguistic study of gender in Ottawa, Ontario". masters. Memorial University of Newfoundland, 2012. URL: https://research.library.mun.ca/2346/ (visited on 04/05/2025).

Annotations: A study of /s/ production in Ottowa, Ontario which includes trans women. One of the sources in my primary prior literature, I refer to its findings that trans women's /s/ production was significantly different from cis people's, and closer to cis women's than to cis men's.

Heffernan: Constructing the "Cisgender Listening Subject"

Dani Heffernan. "Constructing the "Cisgender Listening Subject": Trans-Feminine

Speakers' Commentaries on Voice and Being Heard". PhD thesis. UCLA, 2023. URL: https://escholarship.org/uc/item/5568m53k (visited on 04/06/2025).

Annotations: This study of transfeminine people's metalinguistic commentaries about the voice and being heard emphasizes the importance of the concept of the listening subject (see Calder 2021; Flores and Rosa 2015; Inoue 2003). I use this for its findings on transfeminine people's understanding of the cis gaze.

Holmes-Elliott et al.: The substance of style

Sophie Holmes-Elliott and Erez Levon. "The substance of style: Gender, social class and interactional stance in /s/-fronting in southeast England". In: *Linguistics* 55.5 (Sept. 1, 2017). Publisher: De Gruyter Mouton, pp. 1045–1072. ISSN: 1613-396X. DOI: 10.1515/ling-2017-0020. URL: https://www.degruyterbrill.com/document/doi/10.1515/ling-2017-0020/html (visited on 09/27/2025).

Annotations: This study of /s/ and style reveals style-shifting of /s/ to pertain not only to gender, but aditionally to stance. I take this into consideration in the design of my methodology.

Inoue: The Listening Subject of Japanese Modernity and His Auditory Double

Miyako Inoue. "The Listening Subject of Japanese Modernity and His Auditory Double: Citing, Sighting, and Siting the Modern Japanese Woman". In: *Cultural Anthropology* 18.2 (2003). _eprint: https://anthrosource.onlinelibrary.wiley.com/doi/pdf/10.1525/ pp. 156–193. ISSN: 1548-1360. DOI: 10.1525/can.2003.18.2.156. URL: https://onlinelibrary.wiley.com/doi/abs/10.1525/can.2003.18.2.156 (visited on 09/09/2025).

Annotations: This study of impact of the everpresence of a male listening subject on Japanese women's speech introduces the concept of the listening subject to the study of langauge and gender. I use the concept of the listening subject in my analysis.

Irvine et al.: Language Ideology and Linguistic Differentiation

Judith T. Irvine and Susan Gal. "Language Ideology and Linguistic Differentiation". In: Paul V. Kroskrity. *Regimes of language: ideologies, polities, and identities*. School of American Research advanced seminar series. Santa Fe, NM: School of American Research Press, 2000. ISBN: 978-0-933452-61-9. URL: https://web.stanford.edu/~eckert/PDF/IrvineGal2000.pdf.

Annotations: This paper introduces the semiotic-ideological processes of iconization, fractal recursivity, erasure, which I use as a part of indexicality theory, especially fractal recursivity.

Jessee et al.: The Cisgender Listening Subject in Sociolinguistic Perception

Emmett Jessee and J Calder. "The Cisgender Listening Subject in Sociolinguistic Perception: Transgender Identity Affects Sibilant Categorization in American English". In: *Journal of Sociolinguistics* n/a (n/a Mar. 12, 2025). _eprint: https://onlinelibrary.wiley.com/do ISSN: 1467-9841. DOI: 10.1111/josl.12702. URL: https://onlinelibrary.wiley.com/doi/abs/10.1111/josl.12702 (visited on 04/06/2025).

Annotations: This paper find that cisgender listeners expect transwomen to produce higher frequency /s/ than ciswomen. An important finding as proof that trans voices are subject to greater scrutiny than cis voices when it comes to adherence to gender norms. Also, a good reason for me to focus on /s/! I cite it as emperical proof of the fact that trans voices are subject to greater scrutiny than cis voices.

Livia et al.: "It's a Girl!"

Anna Livia and Kira Hall. ""It's a Girl!": Bringing Perfarmativity Back to Linguistics". In: *Queerly Phrased: Language, Gender, and Sexuality*. Ed. by Anna Livia and Kira Hall. Oxford University Press, Nov. 20, 1997, p. 0. ISBN: 978-0-19-510470-7. DOI: 10.1093/oso/9780195104707.003.0001. URL: https://doi.org/10.1093/oso/9780195104707.003.0001 (visited on 04/08/2025).

Annotations: I reference this introduction's discussion of the history of language and SGS scholarship up to 1997, specifically its discussion of the theoretical ties and tensions between the sociolinguistic literature and other fields including queer theory, feminism, and literary theory.

Maegaard et al.: /s/ Variation and Perceptions of Male Sexuality in Denmark

Marie Maegaard and Nicolai Pharao. "/s/ Variation and Perceptions of Male Sexuality in Denmark". In: *Language, Sexuality, and Power: Studies in Intersectional Sociolinguistics*. _eprint: https://academic.oup.com/book/0/chapter/151100575/chapter-ag-pdf/44978028/book_6883_section_151100575.ag.pdf. Oxford University Press, Dec. 2015. ISBN: 978-0-19-021036-6. DOI: 10.1093/acprof:oso/9780190210366.003.0005. URL: https://doi.org/10.1093/acprof:oso/9780190210366.003.0005.

Annotations: This perception study of /s/ in Danish finds that although fronting is linked to the expected meanings of femininity and gayness when situated in a male voice, these associations do not hold in the context of a prosodic style associated with immigrants, showing the typically-associated meanings of /s/ to be hegemonic and ascribed to to white middle class contexts. I use this paper as an example of research which reveals the typical meaning structure of /s/ to be white.

Parnell-Mooney: Tran/s/gender

James Parnell-Mooney. "Tran/s/gender: assessing the effects of the social con-

struction of gender on speech: a focus on transgender /s/ realisations". MPhil(R). University of Glasgow, 2019. DOI: 10.5525/gla.thesis.41151. URL: https://eleanor.lib.gla.ac.uk/record=b3346280 (visited on 09/26/2025).

Annotations: A study quite similar to my intended methodology, compares transmasculine people's /s/ production across cisgender vs. transgender interlocutors and finds a difference! I cite this as a study I model my own after.

Podesva et al.: /s/exuality in Smalltown California

Robert J. Podesva and Janneke Van Hofwegen. "/s/exuality in Smalltown California: Gender Normativity and the Acoustic Realization of /s/". In: *Language, Sexuality, and Power: Studies in Intersectional Sociolinguistics*. Ed. by Erez Levon and Ronald Beline Mendes. Oxford University Press, Dec. 24, 2015, pp. 168–188. ISBN: 978-0-19-021036-6. DOI: 10.1093/acprof:oso/9780190210366.003.0009. URL: https://doi.org/10.1093/acprof:oso/9780190210366.003.0009 (visited on 04/05/2025).

Annotations: A study of /s/ production in Redding, California, which includes trans women. Similar to Hazenberg (2012), finds significant differences between trans people and cis people, however also finds that trans people's /s/ patterns more with AGAB than gender identity; this is possibly due to the conservative nature of the town. I reference the data the paper finds on trans women's /s/ pronunciations.

Rickford et al.: Addressee-And Topic-Influenced Style Shift

John R Rickford and Faye Mcnair-Knox. "Addressee- And Topic-Influenced Style Shift: A Quantitative Sociolinguistic Study". In: *Sociolinguistic Perspectives On Register*. 1994, pp. 235–276. DOI: 10.1093/oso/9780195083644.003.0011.

Annotations: I cite this early work on AAE style-shifting in my definition of the term 'interlocutor-oriented style-shifting'.

Silverstein: Indexical order and the dialectics of sociolinguistic life

Michael Silverstein. "Indexical order and the dialectics of sociolinguistic life". In: *Language & Communication*. Words and Beyond: Linguistic and Semiotic Studies of Sociocultural Order 23.3 (July 1, 2003), pp. 193–229. ISSN: 0271-5309. DOI: 10.1016/S0271-5309(03)00013-2. URL: https://www.sciencedirect.com/science/article/pii/S0271530903000132 (visited on 04/09/2025).

Annotations: I take up the concept of indexical orders as introduced in this paper.

Zimman: Beyond the cis gays' cis gaze

Lal Zimman. "Beyond the cis gays' cis gaze: The need for a trans linguistics". In: *Gender and Language* 15.3 (Oct. 6, 2021). ISSN: 1747-633X, 1747-6321. DOI: 10.

1558/genl.20883. URL: https://journal.equinoxpub.com/GL/article/view/20883 (visited on 12/19/2024).

Annotations: The 'trans linguistics' that Zimman proposes is a philosophy I take up here. Additionally, I take up the idea of linguistic descriptivism as an apoliticized language ideology.

Zimman: Hegemonic masculinity and the variability of gay-sounding speech

Lal Zimman. "Hegemonic masculinity and the variability of gay-sounding speech: The perceived sexuality of transgender men". In: *Journal of Language and Sexuality* 2.1 (Feb. 27, 2013), pp. 1–39. ISSN: 2211-3770, 2211-3789. DOI: 10.1075/jls.2.1.01zim. URL: http://www.jbe-platform.com/content/journals/10.1075/jls.2.1.01zim (visited on 04/04/2025).

Annotations: This combined production-perception study of /s/, trans men's /s/ behavior is shown to significantly differ from cis men's, but (cis) percievers nonetheless rated them as gay sounding. I'm interested in this data, especially its consideration of the interplay between production and perception, and also interested in its discussion of the meaning structure for /s/ as hegemonic.

Zimman: Transgender Language, Transgender Moment

Lal Zimman. "Transgender Language, Transgender Moment: Toward a Trans Linguistics". In: *The Oxford Handbook of Language and Sexuality*. Ed. by Kira Hall and Rusty Barrett. Oxford University Press, 2020, p. 0. ISBN: 978-0-19-021292-6. DOI: 10.1093/oxfordhb/9780190212926.013.45. URL: https://doi.org/10.1093/oxfordhb/9780190212926.013.45 (visited on 12/19/2024).

Annotations: This paper reviews thirty years of literature on linguistic research on transgender people, as well as other groups which do not fit within the Western gender binary, and proposes 'trans linguistics', a framework which I adopt.